Chapter 3

The Processor

Multiple Controls

- A CPU contains multiple components with different functions, such as **registers, ALU, memory, and PC**.
- These components must be coordinated by different **control signals**.
- Each control signal determines **what a datapath component should do** during instruction execution.

Control target	Control signals	Function
Registers	RegDst, RegWrite	Select the destination register and enable register write
ALU	ALUSrc, ALUOp	Select the ALU input and determine the ALU operation
Data Memory	MemRead, MemWrite	Control memory read and write
Write-back	MemtoReg	Select the value written back to the register
PC / Branch	Branch	Control whether a branch changes the next PC

Different Control Signals

Control signals affect the datapath in different ways.

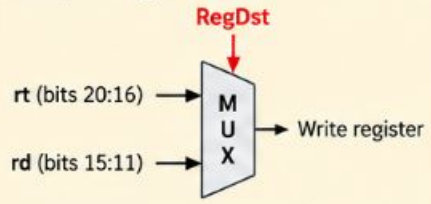
- Some signals select which data path is used.
- Some signals enable or disable an operation.
- Other signals determine ALU or PC behavior.

1. Selection Signals

Select one input of a multiplexer.

Signal	Function
RegDst	Select destination register (rt or rd)
ALUSrc	Select ALU second input (register or immediate)
MemtoReg	Select data to write back to register (ALU result or memory data)

Example: RegDst



2. Enable / Write Signals

Enable or disable an operation.

Signal	Function
RegWrite	Enable register write in the Register File
MemRead	Enable memory read in Data Memory
MemWrite	Enable memory write in Data Memory

Example: RegWrite



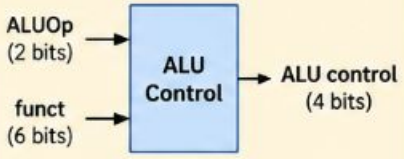
- **RegWrite = 1** → Write the register
- **RegWrite = 0** → Do not write

3. ALU Operation Signal

Specify the required ALU operation class.

Signal	Function
ALUOp (2 bits)	Tell ALU Control which ALU operation is required (e.g., add, subtract, or use funct for R-type).

Example: ALUOp → ALU Control



4. PC / Branch Signal

Participate in determining whether the branch target is selected.

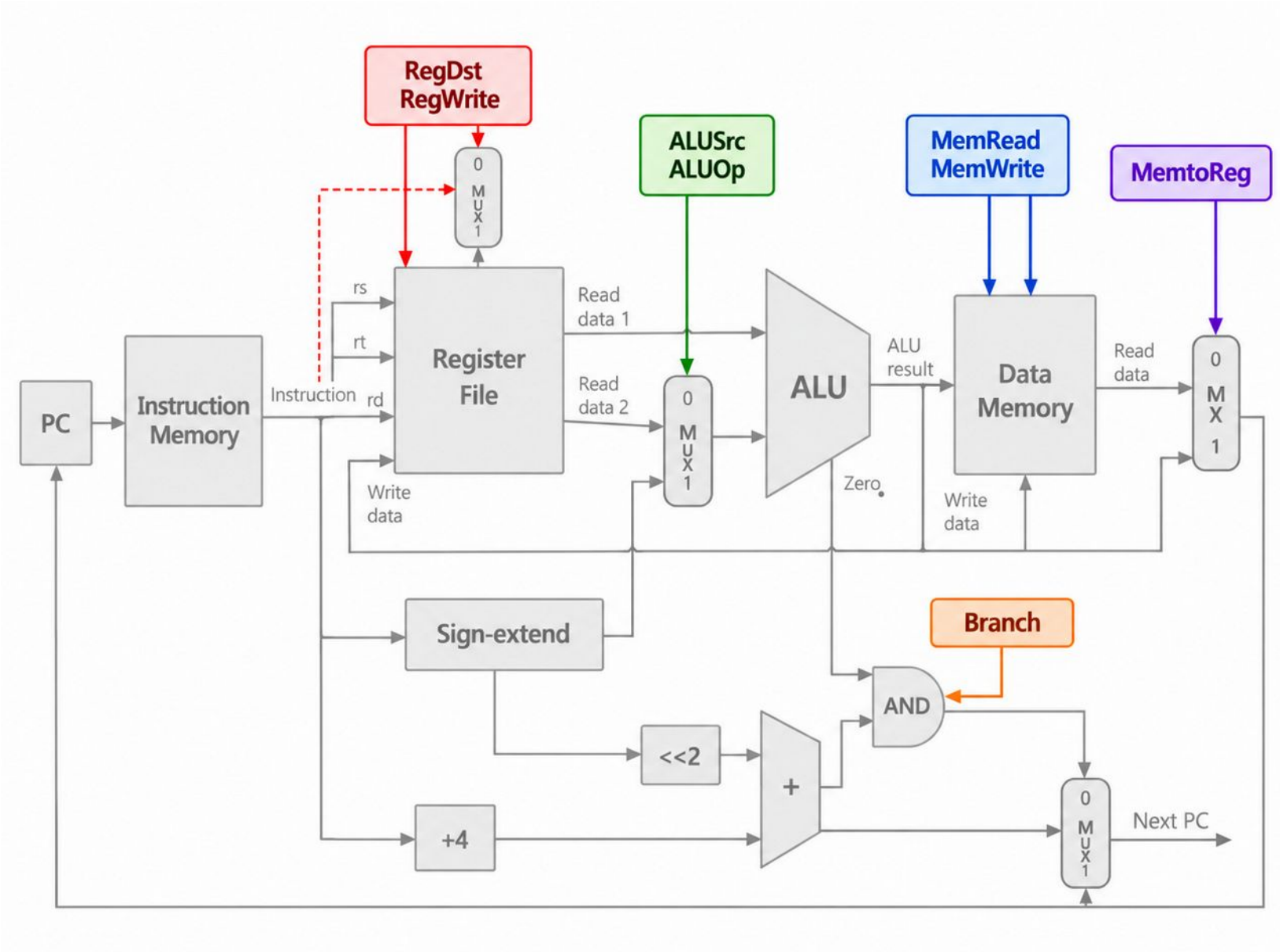
Signal	Function
Branch	Indicate that the current instruction is a branch instruction (e.g., beq).

Example: Branch Control Logic

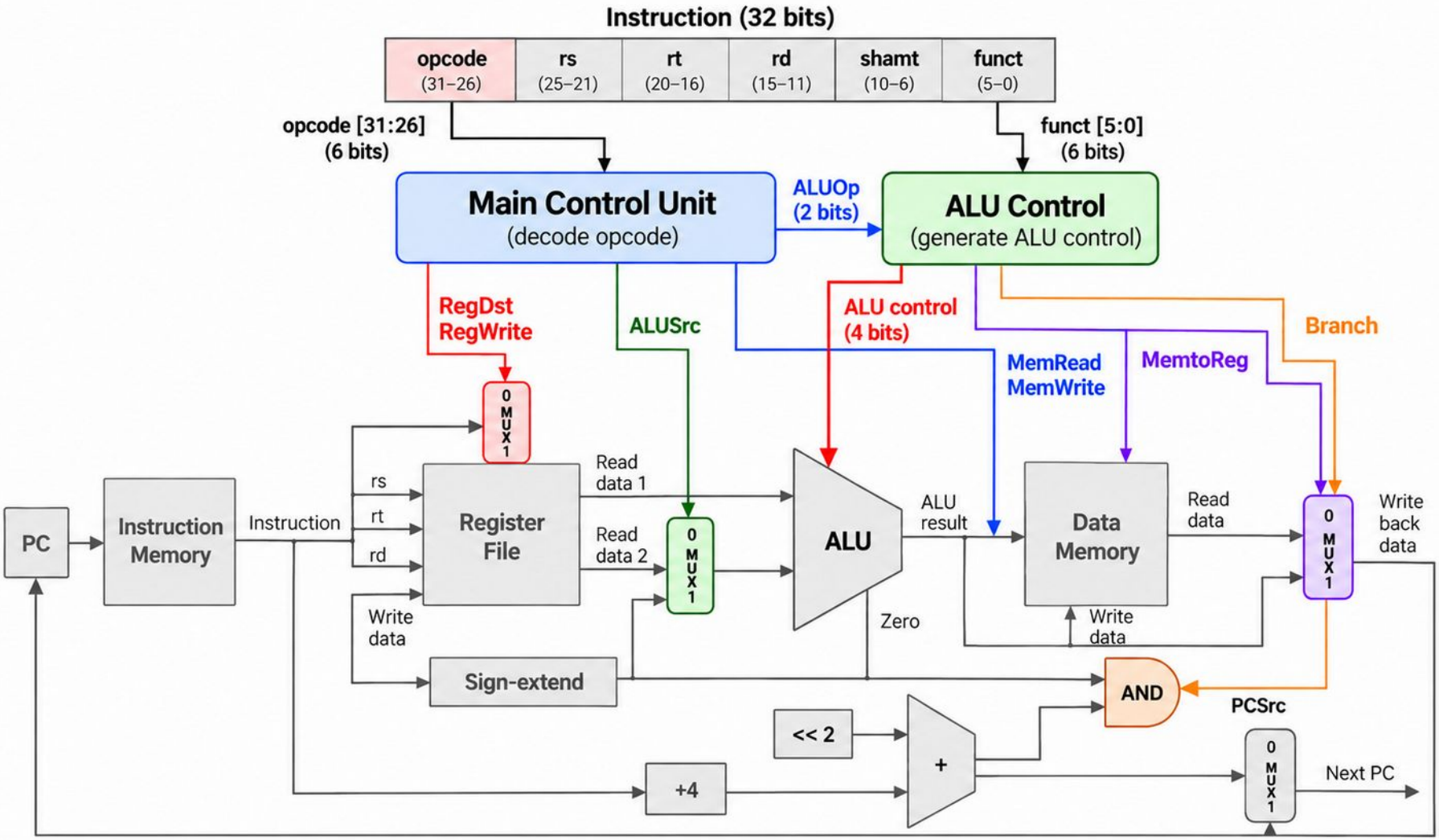


- **Branch = 1** → instruction allows a branch
- **Zero = 1** → ALU comparison result is equal
- **PCSrc = Branch AND Zero**

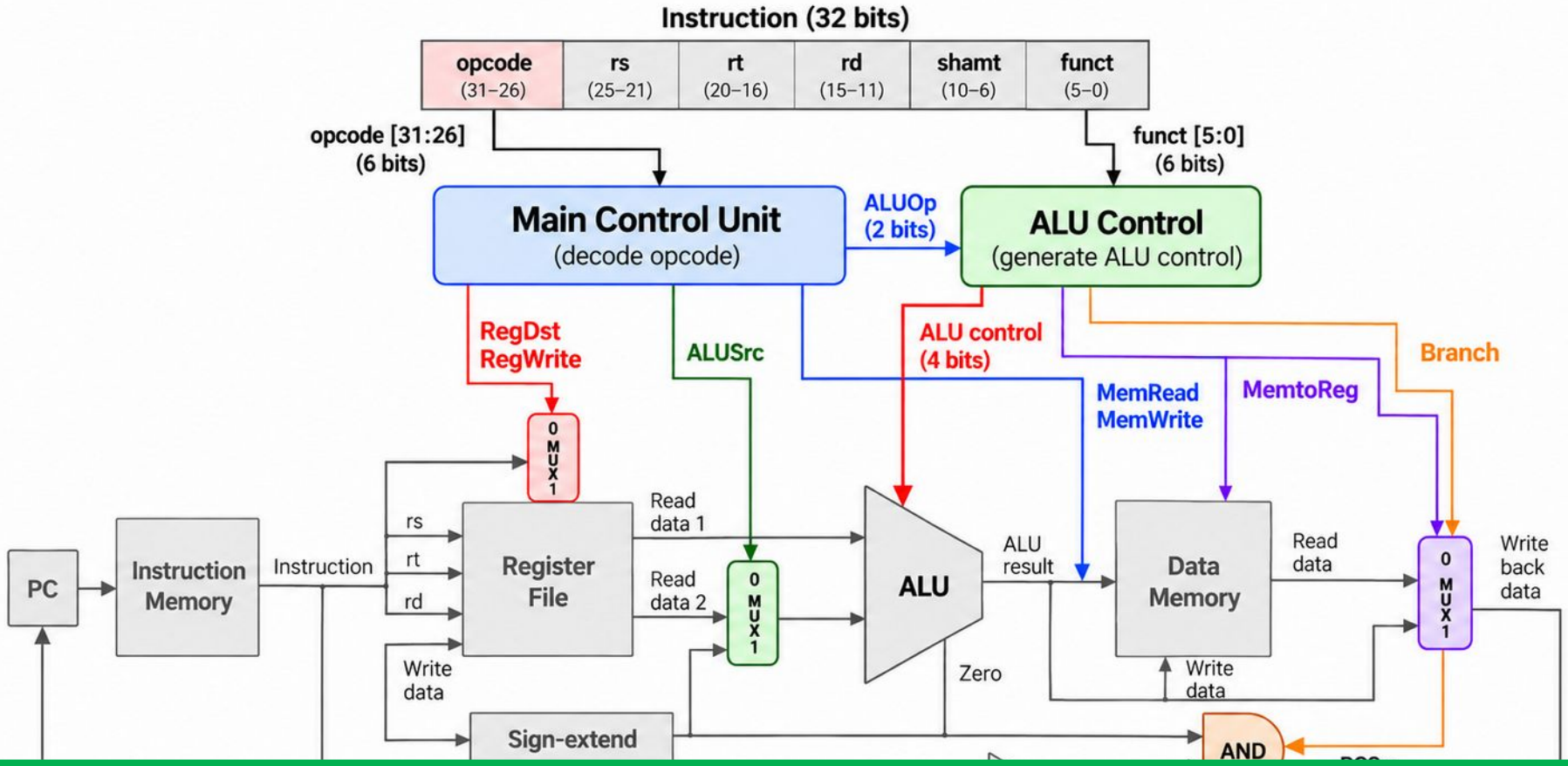
Control Signals in the CPU



Control Signals in the CPU



Control Signals in the CPU

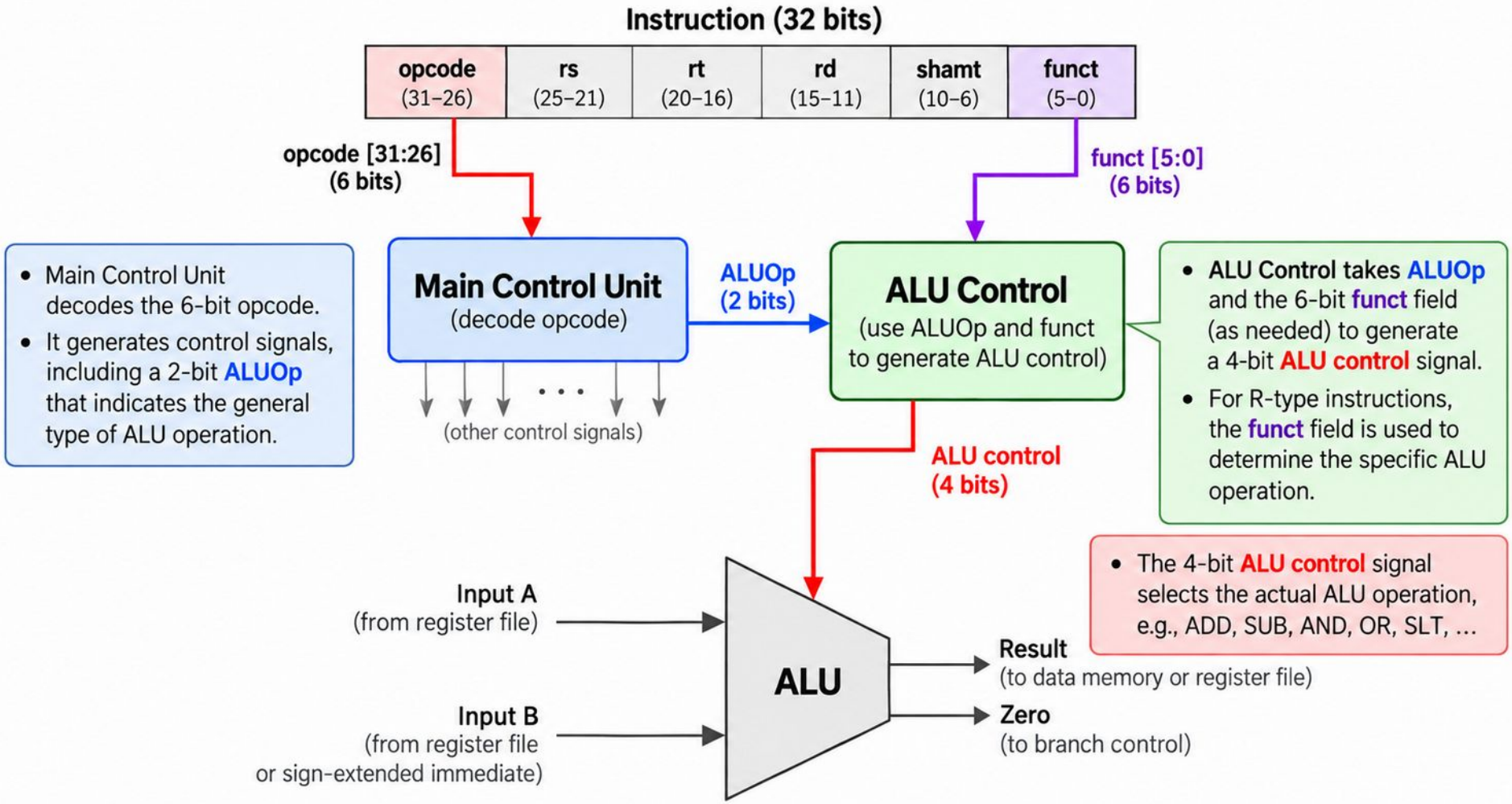


Q1. Does a control signal value of 0 always mean "disabled"

A1.

No. For example, RegDst = 0 selects rt; it does not disable the MUX.

Controlling the ALU



- Main Control Unit decodes the 6-bit opcode.
- It generates control signals, including a 2-bit **ALUOp** that indicates the general type of ALU operation.

- **ALU Control takes ALUOp** and the 6-bit **funct** field (as needed) to generate a 4-bit **ALU control** signal.
- For R-type instructions, the **funct** field is used to determine the specific ALU operation.

- The 4-bit **ALU control** signal selects the actual ALU operation, e.g., ADD, SUB, AND, OR, SLT, ...

ALU Control

- ALU used for
  - Load/Store: F = add
  - Branch: F = subtract
  - R-type: F depends on funct field

ALU control	Function
0000	AND
0001	OR
0010	add
0110	subtract
0111	set-on-less-than
1100	NOR

ALU Control

- ALU used for
  - Load/Store: F = add
  - Branch: F = subtract
  - R-type: F depends on funct field

Q2.
An ALU must perform a SUB operation.
Which 4-bit ALU control signal should be generated? Give two different instruction that may require this operation.
A2.
ALU control = 0110
Two possible situations:
A branch comparison (beq)
An R-type subtract instruction

ALU Control

- Assume 2-bit ALUOp derived from opcode
  - Combinational logic derives ALU control

opcode	ALUOp	Operation	funct	ALU function	ALU control
lw	00	load word	XXXXXX	add	0010
sw	00	store word	XXXXXX	add	0010
beq	01	branch equal	XXXXXX	subtract	0110
R-type	10	add	100000	add	0010
		subtract	100010	subtract	0110
		AND	100100	AND	0000
		OR	100101	OR	0001
		set-on-less-than	101010	set-on-less-than	0111

Understanding Don't Care (X)

- X can be either 0 or 1.
- The input does not affect the required cont rol output.
- X does not mean 0 or an unknown value.

Instruction	ALUOp	funct	ALU Control	Why is funct “Don't Care”?
lw	00	XXXXXX	0010 (ADD)	ALUOp already specifies ADD
sw	00	XXXXXX	0010 (ADD)	ALUOp already specifies ADD
beq	01	XXXXXX	0110 (SUB)	ALUOp already specifies SUB
R-type AND	10	100100	0000 (AND)	funct is required

Understanding Don't Care (X)

- X can be either 0 or 1.

Q3.
Suppose the following 6-bit values are observed:

Instruction	ALUOp	6-bit Value
lw	00	101101
sw	00	011010
beq	01	110001
R-type	10	100101

For which instruction does the 6-bit value affect the ALU Control output?

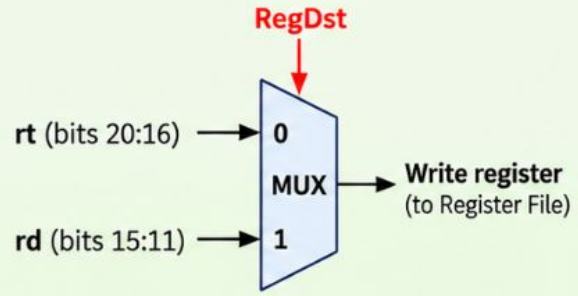
A3.
Only the R-type instruction.
100101 → OR

Control Signals for MUX

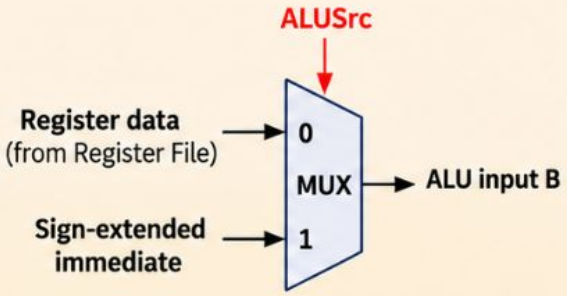
- A multiplexer (MUX) selects one of multiple inputs.
- In the single-cycle datapath, several control signals determine which data path is selected.

Control Signal	0 Selects	1 Selects	Purpose
RegDst	**rt** (bits 20:16)	**rd** (bits 15:11)	Select the destination register
ALUSrc	Register data (from Register File)	Sign-extended immediate	Select the second ALU operand
MemtoReg	ALU result	Memory data (from Data Memory)	Select the value written back to the register

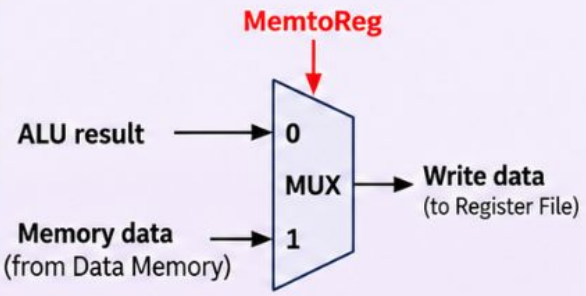
RegDst Select the destination register.



ALUSrc Select the second ALU operand.



MemtoReg Select the value written back to the register.



Control Signals for MUX

- A multiplexer (MUX) selects one of multiple inputs.
- In the single-cycle datapath, several control signals determine which data path is selected

Q4.

Given the following control signals:

Control Signal	Value
RegDst	1
ALUSrc	0
MemtoReg	1

Which input is selected by each MUX?

A4.

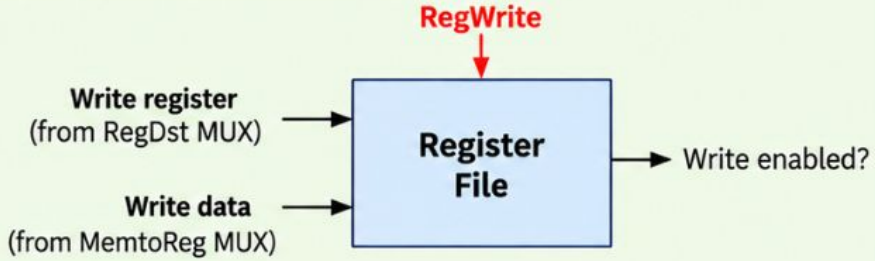
Control Signal	Selected Input
RegDst = 1	rd
ALUSrc = 0	Register data
MemtoReg = 1	Memory data

Control for Reg. & Memory

- Some control signals enable or disable an operation.
- Unlike MUX control signals, 0 and 1 indicate whether an action is performed.

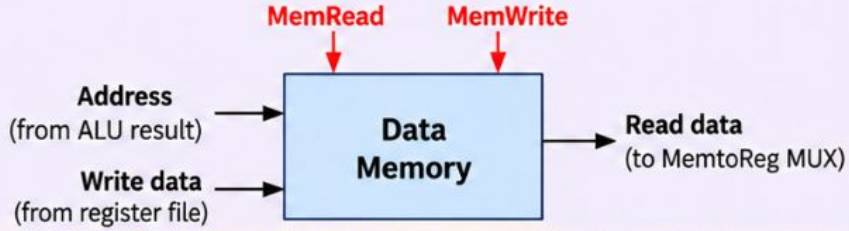
Control Signal	0 (Disabled)	1 (Enabled)	Controlled Component
RegWrite	No register write	Write to register	Register File
MemRead	No memory read	Read from memory	Data Memory
MemWrite	No memory write	Write to memory	Data Memory

RegWrite and the Register File



- **RegWrite** = 1 → Write the register.
- **RegWrite** = 0 → Do not write (register remains unchanged).

MemRead and MemWrite with Data Memory



- **MemRead** = 1 → Read from memory.
- **MemWrite** = 1 → Write to memory.
- **MemRead and MemWrite** should not both be asserted for the instructions considered here.

Control for Reg. & Memory

- Some control signals enable or disable an operation.
- Unlike MUX control signals, 0 and 1 indicate whether an action is performed.

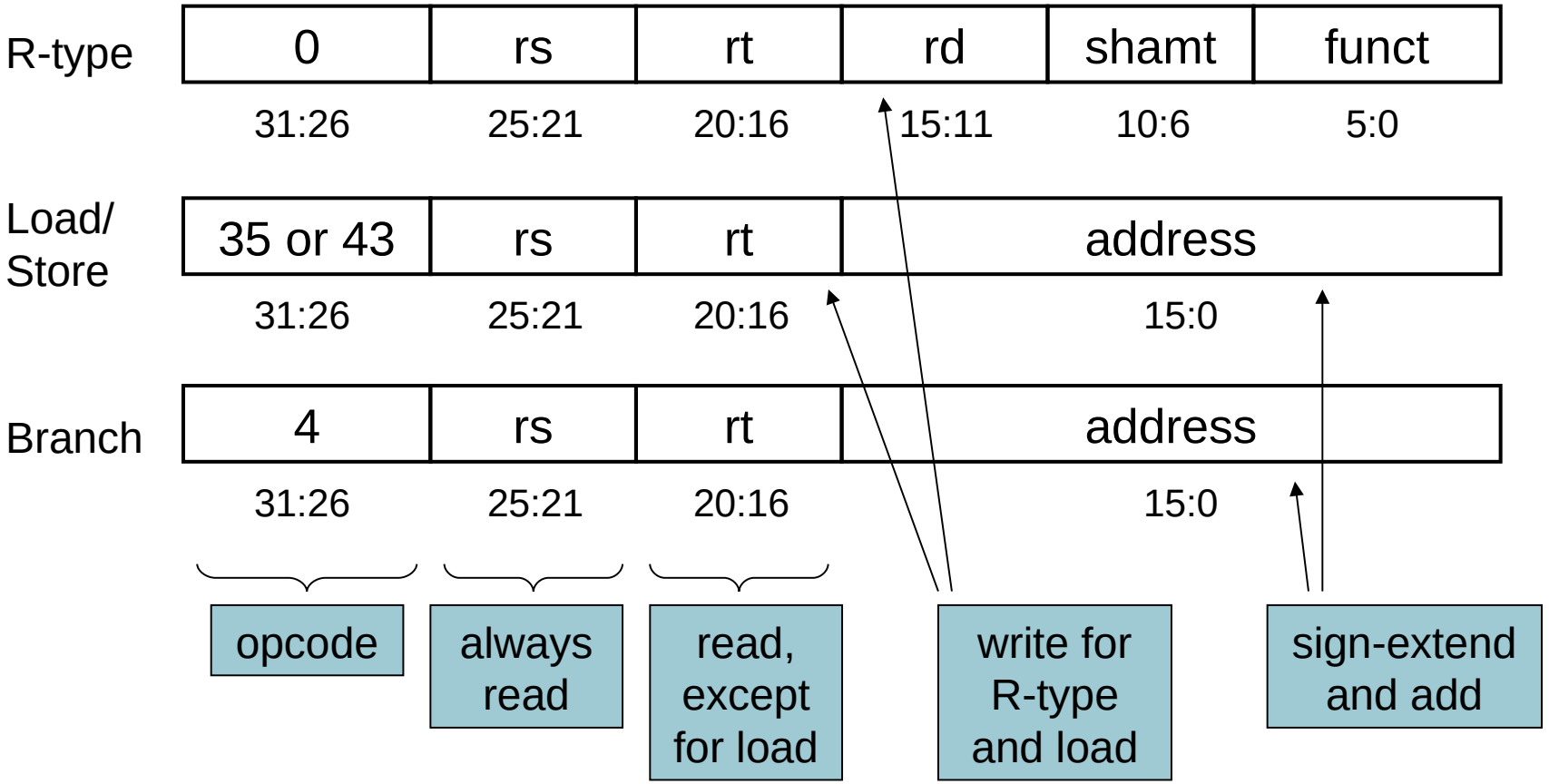
Control Signal	0 (Disabled)	1 (Enabled)	Controlled Component
RegWrite	No register write	Write to register	Register File
MemRead	No memory read	Read from memory	Data Memory
MemWrite	No memory write	Write to memory	Data Memory

Q5.
A control configuration must write a result into the Register File without accessing Data Memory.
What should RegWrite, MemRead, and MemWrite be?

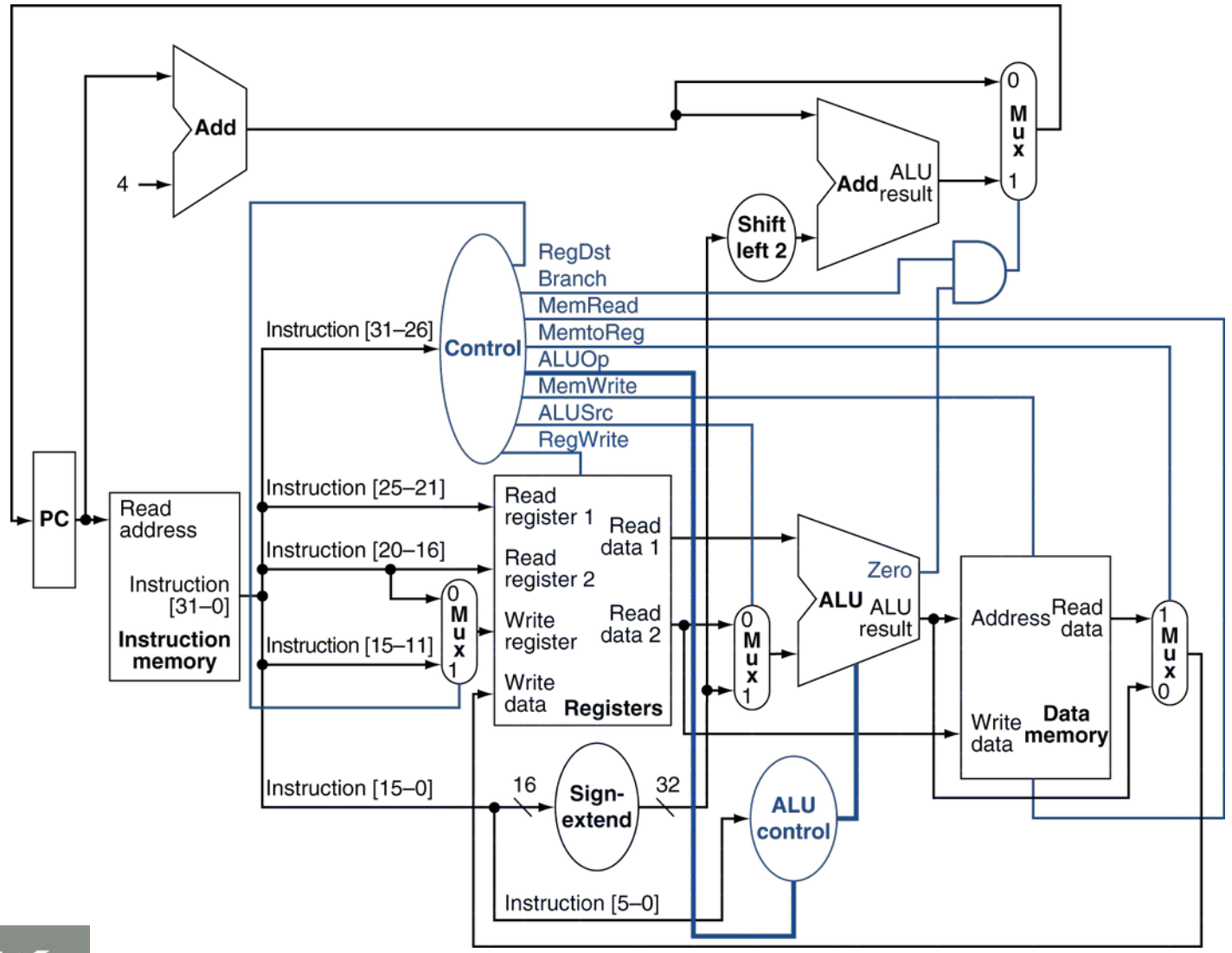
A5.
RegWrite = 1
MemRead = 0
MemWrite = 0

The Main Control Unit

- Control signals derived from instruction



Datapath With Control



Section 3-4 Quick Questions

Q1. Does a control signal value of 0 always mean “disabled”?

Q2. An ALU must perform a SUB operation. Which 4-bit ALU control signal should be generated? Give two different instructions that may require this operation.

Q3. Suppose lw, sw, beq, and R-type have different 6-bit values. For which instruction does the 6-bit value affect the ALU Control output?

Q4. Given RegDst = 1, ALUSrc = 0, and MemtoReg = 1, which input is selected by each MUX?

Q5. A control configuration must write a result into the Register File without accessing Data Memory. What should RegWrite, MemRead, and MemWrite be?

Exercise 1 — Decode the ALU Control

Given the following control information:

ALUOp = 10 funct = 100010 ALU control = 0110

(a) Identify which value is 2-bit, 6-bit, and 4-bit.

(b) Determine the ALU operation.

(c) Determine the corresponding R-type instruction.

Answer

(a) ALUOp: 2-bit; funct: 6-bit; ALU control: 4-bit

(b) ALU operation = SUB

(c) funct = 100010 corresponds to the R-type instruction sub.

Key Point: ALUOp identifies the operation class; funct selects the R-type operation; ALU Control gives the final ALU command.

Exercise 2 — Determine R-Type Control Signals

Consider the instruction: and $t0, $t1, $t2

(a) Determine RegDst, ALUSrc, and RegWrite.

(b) Determine MemRead, MemWrite, and MemtoReg.

(c) Determine ALUOp. Why is the funct field still needed?

Answer

(a) RegDst = 1, ALUSrc = 0, RegWrite = 1

(b) MemRead = 0, MemWrite = 0, MemtoReg = 0

(c) ALUOp = 10. The funct field specifies the exact R-type ALU operation, whi
ch is AND here.

Key Point: R-type instructions use ALUOp = 10, while the funct field specifies the exact ALU operation.

Exercise 3 — Find the Control Error

A student proposes the following control settings for a beq instruction:

ALUSrc = 0 ALUOp = 01 RegWrite = 1

MemRead = 0 MemWrite = 0

(a) Which control value is incorrect?

(b) Correct the value.

(c) Explain what could go wrong if the incorrect value were used.

Answer

(a) RegWrite = 1 is incorrect.

(b) RegWrite should be 0.

(c) beq does not write a result to the Register File. Enabling RegWrite could cause an unintended register write.

The other listed control values are consistent with beq.

Key Point: Check each control signal by asking whether that CPU operation is actually required by the instruction.